

Canonical Tensor Decomposition

Deep Understanding Guide for Questions 1–10

This document expands the section **Canonical Tensor Decomposition** from the uploaded exam guide. It assumes no prior tensor background, introduces every core idea from scratch, and uses the reference list in the guide as the reading map.

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How to Use This Guide

Read the sections in order if the topic is new. Questions 1–6 build the language and theory of CP decomposition: rank, unfoldings, typical rank, border rank, and uniqueness. Questions 7–10 are computational: ALS, compression, Levenberg–Marquardt, and fitting CP models to tensors stored in Tucker or Tensor Train formats.

Main mental model. A CP decomposition tries to represent a multidimensional array as a sum of simple separable patterns. Each simple pattern is a rank-one tensor. The number of patterns is the CP rank.

The attached guide points especially to Kolda–Bader for notation and CP basics, de Silva–Lim for ill-posedness, Kruskal and Sidiropoulos–Bro for uniqueness, Acar–Dunlavy–Kolda for optimization, and Tomasi–Bro for Levenberg–Marquardt and degeneracy in PARAFAC.

Notation and Zero-Knowledge Warm-Up

A vector is a one-index object, a matrix is a two-index object, and a tensor is a multi-index object. For example:

$$\mathbf{x} = (x_i), \quad \mathbf{A} = (a_{ij}), \quad \mathcal{X} = (x_{i_1 i_2 \dots i_d}).$$

Here d is the number of modes, dimensions, or ways. A three-way tensor has entries x_{ijk} . A d -way tensor has entries $x_{i_1 \dots i_d}$.

A rank-one matrix is an outer product of two vectors:

$$\mathbf{A} = \mathbf{a} \mathbf{b}^\top, \quad a_{ij} = a_i b_j.$$

A rank-one tensor is the same idea with more vectors:

$$\mathcal{X} = \mathbf{a}^{(1)} \circ \mathbf{a}^{(2)} \circ \dots \circ \mathbf{a}^{(d)}, \quad x_{i_1 \dots i_d} = a_{i_1}^{(1)} a_{i_2}^{(2)} \dots a_{i_d}^{(d)}.$$

The symbol $\circ$ means outer product. It creates an array by multiplying one entry from each vector.

The CP decomposition is a sum of such rank-one tensors:

$$\mathcal{X} = \sum_{r=1}^R \lambda_r \mathbf{a}_r^{(1)} \circ \mathbf{a}_r^{(2)} \circ \dots \circ \mathbf{a}_r^{(d)}.$$

The factor matrix for mode n collects the mode- n vectors as columns:

$$\mathbf{A}^{(n)} = \begin{bmatrix} \mathbf{a}_1^{(n)} & \mathbf{a}_2^{(n)} & \dots & \mathbf{a}_R^{(n)} \end{bmatrix}.$$

Terminology. CP, CANDECOMP, and PARAFAC refer to essentially the same canonical polyadic representation. The name CP is common in numerical multilinear algebra; PARAFAC is common in chemometrics and factor analysis.

1 Existence of the Canonical Decomposition and Canonical Rank

Question in plain language

When can a tensor be written as a sum of rank-one tensors, and what does it mean to use the smallest possible number of such terms?

Start from zero

For matrices, rank is familiar: a matrix has rank R if it can be written as a sum of R rank-one matrices, but not fewer. CP rank copies this idea for tensors. A rank-one tensor is completely separable across modes. For a three-way tensor, a rank-one term has entries

$$x_{ijk} = a_i b_j c_k.$$

This is simple because the dependence on i , j , and k is separated. A general tensor mixes the indices, so we add several such separable terms:

$$\mathcal{X} = \sum_{r=1}^R \lambda_r \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r.$$

Canonical rank

The canonical rank, or CP rank, is the minimum number of rank-one tensors needed to represent $\mathcal{X}$ exactly:

$$\text{rank}_{\text{CP}}(\mathcal{X}) = \text{the smallest } R \text{ such that } \mathcal{X} \text{ is a sum of } R \text{ rank-one tensors.}$$

For a finite tensor, an exact representation always exists because one can always expand it entry by entry using basis vectors. This entrywise construction may be extremely inefficient, but it proves that some finite rank exists. The real mathematical and computational problem is finding the smallest useful R .

Why tensor rank is harder than matrix rank

For matrices, many good things happen at once:

- the rank is easy to compute with Gaussian elimination or SVD;
- the best rank- R approximation always exists;
- truncated SVD gives the optimal answer in the spectral and Frobenius norms;
- the set of matrices of rank at most R is closed.

For tensors of order three or higher, these properties break. CP rank can be difficult to determine, best low-rank approximation may fail to exist, and algorithms can produce diverging components even when the represented tensor remains bounded.

Key pathology. The exact CP rank exists, but the best approximation with a prescribed smaller rank may not exist. This is not a numerical accident. It is a geometric property of CP rank.

Scaling and permutation ambiguities

A CP decomposition is not unique in a literal column-by-column sense. Even a single rank-one term has scaling ambiguity:

$$\mathbf{a} \circ \mathbf{b} \circ \mathbf{c} = (\alpha \mathbf{a}) \circ (\beta \mathbf{b}) \circ (\gamma \mathbf{c}) \quad \text{whenever} \quad \alpha\beta\gamma = 1.$$

Also, the order of the summands can be permuted. These ambiguities are unavoidable and are considered harmless.

Deep intuition

Think of a tensor as a table whose entries depend on several coordinates. A rank-one term is a pattern that separates the coordinates completely. CP asks: how many separable patterns are needed to build the tensor? The answer is the CP rank. The difficulty is that tensor separability is much more rigid than matrix separability, and sums of separable patterns can approach limits that are not representable with the same number of patterns.

Reference map from the attached guide

- **KB09, pp. 455–469**: notation, outer products, CP decomposition, tensor rank.
- **CC70, pp. 283–292** and **H70, pp. 1–15, 47–58**: classical CANDECOMP/PARAFAC origins.
- **DSL08, pp. 1084–1092**: why best low-rank tensor approximation may fail to exist.

Exam-ready summary

You should be able to define rank-one tensors, write the CP form, define CP rank as a minimum, explain scaling and permutation ambiguity, and state clearly why CP rank differs from matrix rank: bounded CP-rank sets need not be closed, so best low-rank approximations may not exist.

2 Unfolding Matrices and Their Connection with Canonical Rank

Question in plain language

How can a tensor be reshaped into matrices, and what do those matrices tell us about CP rank?

Start from zero

An unfolding, or matricization, is just a systematic reshaping of a tensor into a matrix. For a three-way tensor $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$, the mode-1 unfolding places the first index in the row direction and combines the other two indices in the column direction:

$$\mathbf{X}_{(1)} \in \mathbb{R}^{I \times JK}.$$

Similarly,

$$\mathbf{X}_{(2)} \in \mathbb{R}^{J \times IK}, \quad \mathbf{X}_{(3)} \in \mathbb{R}^{K \times IJ}.$$

The entries are not changed; only their layout is changed.

Matricized CP form

For a three-way CP model

$$\mathcal{X} = \sum_{r=1}^R \lambda_r \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r,$$

define factor matrices

$$\mathbf{A} = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_R], \quad \mathbf{B} = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_R], \quad \mathbf{C} = [\mathbf{c}_1 \ \cdots \ \mathbf{c}_R],$$

and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_R)$. Then the mode-1 unfolding has the form

$$\mathbf{X}_{(1)} = \mathbf{A} \mathbf{\Lambda} (\mathbf{C} \odot \mathbf{B})^\top.$$

Here $\mathbf{C} \odot \mathbf{B}$ is the Khatri–Rao product: it is the columnwise Kronecker product,

$$\mathbf{C} \odot \mathbf{B} = [\mathbf{c}_1 \otimes \mathbf{b}_1 \ \mathbf{c}_2 \otimes \mathbf{b}_2 \ \cdots \ \mathbf{c}_R \otimes \mathbf{b}_R].$$

For a d -way tensor, the mode- n unfolding is

$$\mathbf{X}_{(n)} = \mathbf{A}^{(n)} \mathbf{\Lambda} \left(\mathbf{A}^{(d)} \odot \cdots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \cdots \odot \mathbf{A}^{(1)} \right)^\top.$$

Rank lower bounds

The displayed formula is a product of matrices with at most R columns. Therefore each unfolding has matrix rank at most the CP rank:

$$\text{rank}(\mathbf{X}_{(n)}) \leq R.$$

Since this holds for any CP representation of length R , it implies the useful lower bound

$$\max_n \text{rank}(\mathbf{X}_{(n)}) \leq \text{rank}_{\text{CP}}(\mathcal{X}).$$

Interpretation. If even one unfolding has matrix rank 20, then no CP representation with fewer than 20 rank-one terms can represent the tensor exactly.

Why unfoldings do not solve CP rank completely

Unfolding ranks are only lower bounds. They can underestimate CP rank because matricization destroys some multilinear structure. Two tensors can have the same unfolding ranks but different CP ranks. This is one of the major differences between CP and Tucker/TT decompositions: in Tucker and TT, certain unfolding ranks directly determine the ranks of the format; in CP they do not.

Deep intuition

Unfolding lets us look at a tensor through a matrix lens. CP rank-one terms become rank-one or low-rank contributions in every unfolding. Therefore unfoldings can disprove a proposed small CP rank. But an unfolding compresses several modes into one mega-index, so it forgets which part of a column index came from which original tensor mode. CP rank is sensitive to exactly that mode structure.

Reference map from the attached guide

- **KB09, pp. 459–469:** matricization, n-mode product, Kronecker and Khatri–Rao products, CP in matricized form.
- **DL00, pp. 1256–1264:** unfoldings and multilinear algebra background.
- **TB97, pp. 25–45:** SVD and matrix rank as the matrix-level foundation.

Exam-ready summary

You should be able to define a mode unfolding, write the matricized CP formula, explain the Khatri–Rao product, and state that every unfolding rank is a lower bound for CP rank, but not a complete characterization of it.

3 Arithmetic with Tensors in Canonical Format

Question in plain language

If tensors are stored as CP sums instead of full arrays, how do we add them, multiply them, compute norms, and why do ranks grow?

Start from zero

The full tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times \dots \times n_d}$ has $n_1 n_2 \dots n_d$ entries. CP format stores only factor matrices:

$$\mathbf{A}^{(k)} \in \mathbb{R}^{n_k \times R}, \quad k = 1, \dots, d.$$

If all mode sizes are roughly n , storage is about

$$dnR$$

numbers, much smaller than n^d when R is moderate.

Addition

Suppose

$$\mathcal{X} = \sum_{r=1}^R \lambda_r \bigcirc_{k=1}^d \mathbf{a}_r^{(k)}, \quad \mathcal{Y} = \sum_{s=1}^S \mu_s \bigcirc_{k=1}^d \mathbf{b}_s^{(k)}.$$

Then

$$\mathcal{X} + \mathcal{Y} = \sum_{r=1}^R \lambda_r \bigcirc_{k=1}^d \mathbf{a}_r^{(k)} + \sum_{s=1}^S \mu_s \bigcirc_{k=1}^d \mathbf{b}_s^{(k)}.$$

So the CP rank of the stored representation becomes at most $R + S$. In factor-matrix language, addition concatenates columns:

$$\mathbf{C}^{(k)} = \begin{bmatrix} \mathbf{A}^{(k)} & \mathbf{B}^{(k)} \end{bmatrix}.$$

Scalar product and norm

For two CP tensors, the inner product can be computed without forming the full tensor. For rank-one terms,

$$\left\langle \bigcirc_{k=1}^d \mathbf{a}_r^{(k)}, \bigcirc_{k=1}^d \mathbf{b}_s^{(k)} \right\rangle = \prod_{k=1}^d \left\langle \mathbf{a}_r^{(k)}, \mathbf{b}_s^{(k)} \right\rangle.$$

Therefore

$$\langle \mathcal{X}, \mathcal{Y} \rangle = \sum_{r=1}^R \sum_{s=1}^S \lambda_r \mu_s \prod_{k=1}^d \left(\mathbf{a}_r^{(k)} \right)^\top \mathbf{b}_s^{(k)}.$$

Using Gram matrices $\mathbf{G}^{(k)} = (\mathbf{A}^{(k)})^\top \mathbf{B}^{(k)}$, this becomes a sum over Hadamard products of small matrices. The Frobenius norm follows from

$$\|\mathcal{X}\|_F^2 = \langle \mathcal{X}, \mathcal{X} \rangle.$$

Hadamard product

For entrywise multiplication, rank-one terms multiply mode by mode:

$$\left(\bigcirc_{k=1}^d \mathbf{a}_r^{(k)} \right) * \left(\bigcirc_{k=1}^d \mathbf{b}_s^{(k)} \right) = \bigcirc_{k=1}^d \left(\mathbf{a}_r^{(k)} * \mathbf{b}_s^{(k)} \right).$$

A product of ranks R and S can therefore produce up to RS terms.

Why recompression is unavoidable

Even when the mathematical result has low CP rank, the naive stored representation after addition or multiplication may have many terms. CP arithmetic is therefore often followed by CP compression: find a new CP tensor with fewer terms that approximates the result. This is exactly why questions 8–10 matter.

Rank growth. Addition grows rank additively in the representation. Hadamard products grow rank multiplicatively. Repeated operations can make CP storage inefficient unless compression is applied.

Reference map from the attached guide

- **KB09, pp. 459–473:** tensor operations, CP representation, Khatri–Rao and Kronecker notation, ALS background.
- **O11, pp. 2308–2311:** comparison with TT arithmetic and rank growth.
- **ADK11, pp. 67–76:** objective and gradient computations using factors.

Exam-ready summary

You should know CP storage cost, addition by concatenating components, scalar products through modewise Gram matrices, Hadamard-product rank growth, and the practical need for recompression.

4 Typical Ranks and Generic Rank

Question in plain language

For tensors of a fixed size, what ranks should we expect to see, and why can real tensors have more than one common rank?

Start from zero

For matrices of fixed size $m \times n$, most matrices have rank $\min(m, n)$. Smaller ranks occur in lower-dimensional exceptional sets. Tensor rank behaves differently. For tensors, there may be several ranks that occur on open sets, especially over the real numbers.

Maximal, typical, and generic rank

Let us distinguish three concepts.

- **Maximal rank:** the largest CP rank achieved by any tensor of the given size.
- **Typical rank over the real field:** a rank R is typical if the set of real tensors of rank R contains a nonempty open set.
- **Generic rank over the complex field:** the rank achieved by almost all complex tensors of the given size.

The phrase “almost all” means that the exceptions form a lower-dimensional algebraic set. Over $\mathbb{C}$, there is usually one generic rank. Over $\mathbb{R}$, there can be several typical ranks.

Why the field matters

A real polynomial may have no real root but have complex roots. Similarly, tensor decomposition equations can behave differently over $\mathbb{R}$ and $\mathbb{C}$. A rank that is generic over complex numbers need not describe all open regions of real tensors. This is one reason real tensor rank is subtler than complex tensor rank.

Geometric viewpoint

The set of CP tensors with rank at most R is built from all sums of R rank-one tensors. Its closure is called a secant variety in algebraic geometry. The generic rank is the first R for which this closed family fills the whole ambient tensor space over $\mathbb{C}$. Over $\mathbb{R}$, the same closure can split into regions with different actual ranks.

Intuition. Generic rank answers: what rank should a randomly chosen complex tensor have?
Typical rank answers: what ranks are stable under small real perturbations?

Why matrices do not show the same pathology

For matrices, rank is controlled by determinants of minors. The rank stratification is clean: full-rank matrices form one open dense set. For tensors, the rank-one building blocks are more constrained, and the geometry of sums of rank-one tensors is much more complicated.

Reference map from the attached guide

- **KB09, pp. 470–473:** typical ranks and special tensor-rank behavior.
- **DSL08, pp. 1084–1094:** geometry of rank and low-rank pathologies.
- **K77, pp. 95–104:** ranks of three-dimensional arrays and early estimates.
- **H70, pp. 58–70:** PARAFAC rank in factor models.

Exam-ready summary

You should be able to define maximal rank, typical rank, and generic rank; distinguish real and complex behavior; and explain why tensor rank has multiple stable possibilities while matrix rank has a single full-rank generic situation.

5 A Sequence of Tensors Converging to a Tensor of Higher Rank

Question in plain language

How can tensors of rank R converge to a tensor whose rank is greater than R , and why does this matter for numerical algorithms?

Start from zero

For matrices, if a sequence of matrices all has rank at most R and it converges, the limit also has rank at most R . For tensors, this can fail. A sequence of low-rank tensors can approach a tensor that needs more rank-one terms for an exact representation.

A concrete three-way example

Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^I$, $\mathbf{c}, \mathbf{d} \in \mathbb{R}^J$, and $\mathbf{e}, \mathbf{f} \in \mathbb{R}^K$. Consider

$$\mathcal{X}_\varepsilon = \frac{1}{\varepsilon} \left((\mathbf{a} + \varepsilon \mathbf{b}) \circ (\mathbf{c} + \varepsilon \mathbf{d}) \circ (\mathbf{e} + \varepsilon \mathbf{f}) - \mathbf{a} \circ \mathbf{c} \circ \mathbf{e} \right).$$

For every nonzero ε , $\mathcal{X}_\varepsilon$ is the difference of two rank-one tensors, so its rank is at most 2. Expanding the product gives

$$\mathcal{X}_\varepsilon = \mathbf{b} \circ \mathbf{c} \circ \mathbf{e} + \mathbf{a} \circ \mathbf{d} \circ \mathbf{e} + \mathbf{a} \circ \mathbf{c} \circ \mathbf{f} + \varepsilon n \mathcal{Y}_\varepsilon,$$

where $\mathcal{Y}_\varepsilon$ remains bounded as ε tends to zero. Therefore

$$\mathcal{X}_\varepsilon \longrightarrow \mathcal{X}_0 = \mathbf{b} \circ \mathbf{c} \circ \mathbf{e} + \mathbf{a} \circ \mathbf{d} \circ \mathbf{e} + \mathbf{a} \circ \mathbf{c} \circ \mathbf{f}.$$

The limit is a sum of three rank-one tensors. For suitable independent choices of the vectors, its rank is 3 even though it is the limit of rank-at-most-2 tensors.

Border rank

The border rank of $\mathcal{X}$ is the smallest R such that $\mathcal{X}$ can be approximated arbitrarily well by rank- R tensors:

$$\text{rank}_{\text{border}}(\mathcal{X}) = \text{extthesmallest } R \text{ such that } \mathcal{X} \in \overline{\{\mathcal{Y} : \text{rank}_{\text{CP}}(\mathcal{Y}) \leq R\}}.$$

In the example, the limit tensor can have ordinary rank 3 but border rank 2.

Why algorithms suffer

If we try to find the best rank-2 approximation to $\mathcal{X}_0$, a minimizing sequence may push ε closer and closer to zero. The two rank-one components then become very large and nearly cancel. Numerically, this appears as degeneracy: factor norms blow up while the represented tensor stays bounded.

Practical warning. Diverging CP components are often not just bad scaling. They can indicate that the requested CP rank is below what is needed for an attainable best approximation.

Deep intuition

The expression above behaves like a derivative of a rank-one curve. A derivative of rank-one tensors need not itself be rank one or even rank two. This is the geometric reason that limits of low-rank tensors can have higher ordinary rank.

Reference map from the attached guide

- **DSL08, pp. 1084–1101:** explicit examples and ill-posedness of best low-rank tensor approximation.
- **KB09, pp. 470–473:** border rank and CP-rank pathologies.
- **TB05, pp. 163–170:** practical degeneracy in PARAFAC algorithms.
- **K77, pp. 95–102:** foundational definitions for three-way arrays.

Exam-ready summary

You should be able to explain border rank, give the idea of a rank-2 sequence converging to a rank-3 tensor, and connect this to non-closedness, nonexistence of best low-rank approximations, and diverging components in CP algorithms.

6 Kruskal Ranks and the d -Dimensional Generalization of Kruskal's Theorem

Question in plain language

When is a CP decomposition essentially unique, and how can column independence of factor matrices guarantee uniqueness?

Start from zero

Matrix factorizations are often nonunique. For example, $\mathbf{A} = \mathbf{U}\mathbf{V}^T$ can be modified by inserting $\mathbf{M}\mathbf{M}^{-1}$. CP decomposition is surprisingly different: under certain conditions, the factors can be uniquely determined up to scaling and permutation.

Kruskal rank

The Kruskal rank, or k -rank, of a matrix $\mathbf{A}$ is the largest integer k_A such that every set of k_A columns of $\mathbf{A}$ is linearly independent.

$$k_A = \text{extlargest } k \text{ such that any } k \text{ columns of } \mathbf{A} \text{ are independent.}$$

This is stronger than ordinary rank. Ordinary rank only says that at least one group of k columns can be independent. Kruskal rank says every group of that size is independent.

Three-mode Kruskal theorem

For a three-way CP decomposition with R components and factor matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, a classical sufficient condition for essential uniqueness is

$$k_A + k_B + k_C \geq 2R + 2.$$

Essential uniqueness means that any other CP decomposition with R terms represents the same components, except for column permutation and modewise scaling whose product per component is one.

Generalization to d modes

For a d -way tensor with factor matrices $\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(d)}$, a common Kruskal-style sufficient condition is

$$\sum_{n=1}^d k_n \geq 2R + d - 1,$$

where k_n is the Kruskal rank of $\mathbf{A}^{(n)}$. For $d = 3$, this reduces to the classical condition.

Why this condition makes sense

A CP decomposition could fail to be unique if one can mix columns across components without changing the tensor. High Kruskal ranks prevent such mixing because every small subset of columns is independent. The condition says that, collectively across modes, the factors are independent enough to identify the components.

Uniqueness is a major CP advantage. In applications, each rank-one component may correspond to a latent source or factor. Kruskal's theorem gives conditions under which those sources are identifiable.

Limits of the theorem

Kruskal's condition is sufficient, not necessary. A decomposition may be unique even if the inequality fails. Also, computing exact Kruskal rank can be hard because it requires checking many column subsets.

Reference map from the attached guide

- **K77, pp. 95–121:** rank, k-rank, uniqueness problem, and proof idea.
- **SB00, pp. 229–239:** N-way/generalized uniqueness conditions.
- **KB09, pp. 469–473:** modern presentation of CP uniqueness and Kruskal condition.

Exam-ready summary

You should define Kruskal rank, state the three-mode condition $k_A + k_B + k_C \geq 2R + 2$, explain scaling and permutation ambiguities, state the d -mode generalization, and emphasize that Kruskal conditions are sufficient but not necessary.

7 ALS for Approximating a Tensor by a Tensor in Canonical Format

Question in plain language

How does alternating least squares compute a CP approximation when the full tensor is given?

Start from zero

Suppose we have a tensor $\mathcal{X}$ and want a rank- R CP approximation

$$\hat{\mathcal{X}} = \sum_{r=1}^R \lambda_r \bigcirc_{n=1}^d \mathbf{a}_r^{(n)}.$$

This is a nonlinear least-squares problem:

$$\min_{\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(d)}} \frac{1}{2} \left\| \mathcal{X} - \hat{\mathcal{X}} \right\|_{\text{F}}^2.$$

It is nonlinear because the factors from different modes multiply each other. ALS makes it easy by fixing all factor matrices except one. With all other factors fixed, the problem becomes an ordinary linear least-squares problem.

One ALS update

For mode n , the matricized model is

$$\hat{\mathbf{X}}_{(n)} = \mathbf{A}^{(n)} \mathbf{Z}^{(n)\top},$$

where

$$\mathbf{Z}^{(n)} = \mathbf{A}^{(d)} \odot \dots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \dots \odot \mathbf{A}^{(1)}.$$

The least-squares update is

$$\mathbf{A}^{(n)} = \mathbf{X}_{(n)} \mathbf{Z}^{(n)} \left(\mathbf{Z}^{(n)\top} \mathbf{Z}^{(n)} \right)^{-1}.$$

The product $\mathbf{X}_{(n)} \mathbf{Z}^{(n)}$ is called MTTKRP: matricized tensor times Khatri–Rao product. It is the computational core of CP-ALS.

Efficient normal equations

The Gram matrix of the Khatri–Rao product can be computed from small Gram matrices:

$$\mathbf{Z}^{(n)\top} \mathbf{Z}^{(n)} = \underset{m \neq n}{*} \left(\mathbf{A}^{(m)\top} \mathbf{A}^{(m)} \right),$$

where $*$ denotes entrywise multiplication of the $R \times R$ Gram matrices. This avoids forming huge matrices when possible.

The ALS cycle

A typical ALS iteration is:

1. initialize all factor matrices;
2. update $\mathbf{A}^{(1)}$ while holding the others fixed;
3. update $\mathbf{A}^{(2)}$, then $\mathbf{A}^{(3)}$, and so on;
4. normalize columns and store weights λ_r ;
5. check convergence using fit, relative change, gradient norm, or iteration limit.

Why ALS is popular

ALS is simple, uses familiar least-squares steps, and often works well. Its main cost is repeated MTTKRP. It is also easy to adapt to sparse tensors because MTTKRP can be computed from nonzero entries.

Why ALS can fail or slow down

ALS solves a nonconvex problem. It can converge to local minima, saddle points, or degenerate solutions. It can also be very slow when factors are nearly collinear, because each one-mode least-squares problem becomes ill-conditioned.

ALS is not globally optimal. Each subproblem is solved optimally, but the full CP objective is nonconvex. Good initialization, normalization, and diagnostics matter.

Reference map from the attached guide

- **KB09, pp. 464–473:** CP model, matricized CP, ALS algorithm.
- **ADK11, pp. 67–76:** optimization formulation, gradient, MTTKRP.
- **ADKM11, pp. 41–48:** weighted CP and incomplete data.
- **TB97, pp. 84–107:** least squares and normal equations.

Exam-ready summary

You should derive the ALS update from the matricized CP form, define MTTKRP, explain the Gram/Hadamard identity for the normal equations, and discuss normalization, stopping criteria, nonconvexity, degeneracy, and slow convergence.

8 ALS for Approximating a CP Tensor by a Lower-Rank Tensor

Question in plain language

If a tensor is already stored as a high-rank CP decomposition, how can ALS compress it to a lower CP rank without forming the full tensor?

Start from zero

Suppose the source tensor is already in CP format:

$$\mathcal{X} = \sum_{p=1}^P \alpha_p \bigcirc_{n=1}^d \mathbf{u}_p^{(n)}.$$

We want a lower-rank CP approximation

$$\mathcal{Y} = \sum_{r=1}^R \lambda_r \bigcirc_{n=1}^d \mathbf{a}_r^{(n)}, \quad R < P.$$

The optimization problem is

$$\min_{\mathcal{Y} \in \text{CP}_R} \frac{1}{2} \|\mathcal{X} - \mathcal{Y}\|_{\text{F}}^2.$$

The full tensor may be enormous, so the key is to compute ALS quantities using only the CP factors of $\mathcal{X}$ and $\mathcal{Y}$.

Objective through Gram matrices

The squared error expands as

$$\|\mathcal{X} - \mathcal{Y}\|_{\text{F}}^2 = \|\mathcal{X}\|_{\text{F}}^2 - 2\langle \mathcal{X}, \mathcal{Y} \rangle + \|\mathcal{Y}\|_{\text{F}}^2.$$

Each term can be computed through modewise Gram matrices. For example,

$$\langle \mathcal{X}, \mathcal{Y} \rangle = \sum_{p=1}^P \sum_{r=1}^R \alpha_p \lambda_r \prod_{n=1}^d \left(\mathbf{u}_p^{(n)} \right)^{\text{T}} \mathbf{a}_r^{(n)}.$$

No full tensor entries are needed.

ALS update without expansion

The usual ALS update for $\mathbf{A}^{(n)}$ needs $\mathbf{X}_{(n)} \mathbf{Z}^{(n)}$. Since $\mathcal{X}$ is CP, this product can be computed from source and target factors. The source unfolding has

$$\mathbf{X}_{(n)} = \mathbf{U}^{(n)} \mathbf{D}_{\alpha} \left(\mathbf{U}^{(d)} \odot \dots \odot \mathbf{U}^{(n+1)} \odot \mathbf{U}^{(n-1)} \odot \dots \odot \mathbf{U}^{(1)} \right)^{\text{T}}.$$

Multiplying by the target Khatri–Rao product gives products of small cross-Gram matrices:

$$\left(\mathbf{U}^{(m)} \right)^{\text{T}} \mathbf{A}^{(m)}, \quad m \neq n.$$

Thus the update can be implemented by contractions among factor matrices, not by expanding $\mathcal{X}$.

Why the problem remains hard

Even though the source is already compact, reducing CP rank is still nonlinear and nonconvex. A lower-rank CP approximation may fail to exist for the requested R . Even when it exists, ALS may find a local minimum.

When CP compression is used

CP compression appears after arithmetic operations, after elementwise products, after adding many CP tensors, or as a postprocessing step to simplify a model. The goal is not just smaller storage; it can also improve interpretability by reducing the number of components.

Matrix analogy. This resembles compressing a high-rank matrix approximation by SVD truncation, but CP has no universally optimal truncated SVD. We must solve a nonconvex optimization problem.

Reference map from the attached guide

- **KB09, pp. 464–473:** CP representation, ALS, Gram and Khatri–Rao notation.
- **ADK11, pp. 67–78:** nonlinear least-squares formulation and computations without full tensors.
- **HMT11, pp. 217–226:** matrix analogy for low-rank compression.
- **O11, pp. 2308–2311:** contrast with stable rounding in TT format.

Exam-ready summary

You should explain CP compression, write the error norm expansion, show how scalar products use Gram matrices, describe how ALS updates avoid full tensor expansion, and emphasize that CP compression remains nonlinear, nonconvex, and sometimes ill-posed.

9 Levenberg–Marquardt Method for CP Approximation and CP Compression

Question in plain language

How does Levenberg–Marquardt solve CP approximation as a nonlinear least-squares problem, and why can it be more robust than ALS?

Start from zero

CP approximation minimizes a residual. Collect all factor entries into one parameter vector θ . The model tensor is $\hat{\mathcal{X}}(\theta)$, and the residual is

$$\mathbf{r}(\theta) = \text{vec} \left(\hat{\mathcal{X}}(\theta) - \mathcal{X} \right).$$

The objective is

$$\min_{\theta} \frac{1}{2} \|\mathbf{r}(\theta)\|_2^2.$$

This is a nonlinear least-squares problem because the entries of $\hat{\mathcal{X}}$ multiply factors from different modes.

Gauss–Newton idea

Near the current point, linearize the residual:

$$\mathbf{r}(\theta + \Delta) \approx \mathbf{r}(\theta) + \mathbf{J}\Delta,$$

where $\mathbf{J}$ is the Jacobian matrix of residual derivatives. Gauss–Newton chooses Δ by solving

$$\mathbf{J}^\top \mathbf{J} \Delta = -\mathbf{J}^\top \mathbf{r}.$$

This can converge faster than ALS when the local linear model is accurate.

Levenberg–Marquardt damping

Levenberg–Marquardt adds a damping term:

$$\left(\mathbf{J}^\top \mathbf{J} + \mu \mathbf{I} \right) \Delta = -\mathbf{J}^\top \mathbf{r}.$$

The parameter μ controls the behavior:

- small μ : step behaves like Gauss–Newton, often fast near a good solution;
- large μ : step behaves more like gradient descent, often safer when far from the solution or when the normal equations are ill-conditioned.

Block structure for CP

The CP Jacobian has blocks corresponding to factor matrices. Derivatives with respect to one factor matrix resemble Khatri–Rao products of the other factor matrices. This structure means $\mathbf{J}$ is usually not formed as a huge dense matrix. Instead, implementations compute products with $\mathbf{J}$ and $\mathbf{J}^\top$, or exploit approximate Hessian blocks.

LM for compression

For CP compression, the residual is the difference between a high-rank source CP tensor and a lower-rank target CP tensor. All inner products and Jacobian-related quantities can again be computed through factor contractions and Gram matrices. The same nonlinear least-squares method applies.

ALS versus LM

Aspect	ALS	Levenberg–Marquardt
Basic step	Solve one factor at a time	Update all parameters together
Problem view	Alternating linear least squares	Damped nonlinear least squares
Strength	Simple, cheap iterations	Can handle coupling between modes better
Weakness	Slow for collinear factors; local minima	More complex and potentially expensive
Key computation	MTTKRP and small normal equations	Jacobian or Jacobian-product structure

Damping is not decoration. CP problems are often ill-conditioned. The damping term stabilizes steps when the Gauss–Newton system is nearly singular or when the local model is unreliable.

Reference map from the attached guide

- **TB97, pp. 84–107:** least squares and normal equations.
- **KB09, pp. 464–473:** CP objective and ALS foundation.
- **ADK11, pp. 67–86:** scalable optimization, gradients, and Hessian approximations.
- **TB05, pp. 163–180:** Levenberg–Marquardt for PARAFAC with missing values.

Exam-ready summary

You should define residual, Jacobian, Gauss–Newton system, LM damping, and explain the role of CP block structure. You should also contrast ALS and LM: ALS updates one mode at a time; LM updates coupled parameters using a damped nonlinear least-squares step.

10 ALS and Levenberg–Marquardt for Approximating Tucker and TT Tensors by a CP Tensor

Question in plain language

If the source tensor is not stored explicitly but in Tucker or Tensor Train form, how can we fit a CP tensor to it using ALS or Levenberg–Marquardt?

Start from zero

Sometimes the tensor $\mathcal{X}$ is too large to form explicitly. It may be stored in another compressed format.

In Tucker format,

$$\mathcal{X} = \mathcal{G} \times_1 \mathbf{U}^{(1)} \times_2 \mathbf{U}^{(2)} \cdots \times_d \mathbf{U}^{(d)},$$

where $\mathcal{G}$ is a smaller core and the $\mathbf{U}^{(n)}$ are factor matrices.

In Tensor Train format, an entry is represented by multiplying slices of cores:

$$x_{i_1 \dots i_d} = \mathbf{G}_1(i_1) \mathbf{G}_2(i_2) \cdots \mathbf{G}_d(i_d),$$

with boundary ranks equal to one.

The goal is to approximate $\mathcal{X}$ by a CP tensor $\hat{\mathcal{X}}$ without expanding $\mathcal{X}$.

What ALS needs

A CP-ALS update needs MTTKRP:

$$\mathbf{X}_{(n)} \mathbf{Z}^{(n)}.$$

When $\mathcal{X}$ is Tucker or TT, this product must be computed by contractions with the compressed cores.

For Tucker, the contractions pass through the small core. If the target CP factors are $\mathbf{A}^{(m)}$, then modewise products such as

$$\left(\mathbf{U}^{(m)} \right)^\top \mathbf{A}^{(m)}$$

compress the target factors into Tucker-core coordinates. The expensive full tensor is never formed.

For TT, MTTKRP is computed by sweeping contractions through the TT cores, leaving one physical mode open. The cost depends on TT ranks, CP rank, mode sizes, and the number of modes.

What LM needs

Levenberg–Marquardt needs the residual norm, gradient, and Jacobian-related products. These can also be computed by contractions:

$$\|\mathcal{X} - \hat{\mathcal{X}}\|_{\mathbb{F}}^2 = \|\mathcal{X}\|_{\mathbb{F}}^2 - 2\langle \mathcal{X}, \hat{\mathcal{X}} \rangle + \|\hat{\mathcal{X}}\|_{\mathbb{F}}^2.$$

The term $\|\hat{\mathcal{X}}\|_{\mathbb{F}}^2$ is CP-only and uses Gram matrices. The term $\|\mathcal{X}\|_{\mathbb{F}}^2$ is computed inside Tucker or TT format. The cross term $\langle \mathcal{X}, \hat{\mathcal{X}} \rangle$ is computed by contracting Tucker or TT cores with CP factors.

Why convert Tucker or TT to CP?

Tucker and TT are often more stable for storage and rounding, while CP can be more interpretable and cheaper when the rank is small. Converting to CP may be useful when:

- one wants separable rank-one components for interpretation;

- subsequent operations are easiest in CP form;
- a lower-rank CP model gives a compact summary;
- CP factors are required by a downstream method.

Cost drivers

The main parameters controlling the cost are:

- number of modes d ;
- mode sizes $n_1, \dots, n_d$;
- target CP rank R ;
- Tucker multilinear ranks or TT ranks;
- number of ALS or LM iterations;
- cost of contractions needed for MTTKRP, gradient, or Jacobian products.

Deep intuition

The algorithm asks the source format questions of the form: “how does this tensor interact with a rank-one or CP test tensor?” Tucker and TT formats can answer these questions through contractions. CP fitting therefore becomes a problem of repeatedly querying the compressed source in exactly the ways required by ALS or LM.

Reference map from the attached guide

- **KB09, pp. 464–473:** CP objective, ALS, and Khatri–Rao notation.
- **KB09, pp. 473–480:** Tucker model and matricized forms.
- **O11, pp. 2297–2311:** TT cores, unfoldings, rounding, and operations.
- **ADK11, pp. 67–86:** nonlinear optimization viewpoint useful for LM-style CP fitting.

Exam-ready summary

You should explain that ALS and LM need objective values, MTTKRP, gradients, and Jacobian-related quantities. When the source is Tucker or TT, these are computed by contractions with the compact cores, never by forming the full tensor. You should identify the cost parameters and motivate why CP conversion may be useful.

Cross-Question Concept Map

Concept	Meaning and where it appears
Rank-one tensor CP rank	A fully separable outer product. Foundation of Questions 1–3. Minimum number of rank-one terms. Question 1; pathologies in Questions 4–5.
Unfolding	Matrix view of a tensor. Gives lower bounds for CP rank in Question 2 and supports ALS in Question 7.
Khatri–Rao product	Columnwise Kronecker product. Central in matricized CP formulas and ALS updates.
Gram matrix	Matrix of pairwise factor inner products. Used for scalar products, norms, compression, ALS, and LM.
Typical rank Border rank	Rank that occurs on an open set of real tensors. Question 4. Rank visible through limits. Explains non-closedness and ill-posed approximation in Question 5.
Kruskal rank	Strong column independence measure. Used for uniqueness in Question 6.
MTTKRP	Matricized tensor times Khatri–Rao product. Main computational kernel of CP-ALS in Question 7.
Damping Contraction	Stabilization in Levenberg–Marquardt. Central in Question 9. Summing over shared indices. The mechanism that lets us fit CP models to Tucker or TT tensors in Question 10.

Common Exam Pitfalls

1. Saying that CP rank behaves like matrix rank. It does not: low-rank CP sets may be non-closed.
2. Confusing CP rank with Tucker rank. CP rank is a single integer; Tucker rank is a vector of unfolding ranks.
3. Treating unfolding ranks as equal to CP rank. They are lower bounds, not exact characterizations.
4. Forgetting scaling and permutation ambiguities in uniqueness statements.
5. Presenting Kruskal’s condition as necessary. It is sufficient.
6. Explaining ALS as globally optimal. It is only locally alternating least squares.
7. Ignoring degeneracy when a requested low rank is too small.
8. Forming full tensors in compression problems where factor contractions are intended.
9. Treating LM as just “ALS with damping.” LM is a coupled nonlinear least-squares method.
10. Forgetting that Tucker/TT-to-CP fitting is driven by contractions, MTTKRP, gradients, and residual norms.

Reference Map Used

The following abbreviations and page ranges are taken from the uploaded exam guide and used as the reading map for this PDF.

Tag	Source
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ADK11	Acar, Dunlavy, and Kolda, scalable optimization for fitting canonical tensor decompositions, <i>Journal of Chemometrics</i> 25(2), 67–86, 2011.
ADKM11	Acar, Dunlavy, Kolda, and Mørup, scalable tensor factorizations for incomplete data, <i>Chemometrics and Intelligent Laboratory Systems</i> 106(1), 41–56, 2011.
CC70	Carroll and Chang, CANDECOMP model for individual differences in multidimensional scaling, <i>Psychometrika</i> 35, 283–319, 1970.
DL00	De Lathauwer, De Moor, and Vandewalle, multilinear singular value decomposition, <i>SIAM Journal on Matrix Analysis and Applications</i> 21(4), 1253–1278, 2000.
DSL08	de Silva and Lim, tensor rank and ill-posedness of the best low-rank approximation problem, <i>SIAM Journal on Matrix Analysis and Applications</i> 30(3), 1084–1127, 2008.
H70	Harshman, foundations of the PARAFAC procedure, <i>UCLA Working Papers in Phonetics</i> 16, 1–84, 1970.
HMT11	Halko, Martinsson, and Tropp, randomized algorithms for low-rank matrix decompositions, <i>SIAM Review</i> 53(2), 217–288, 2011.
K77	Kruskal, three-way arrays: rank and uniqueness of trilinear decompositions, <i>Linear Algebra and Its Applications</i> 18(2), 95–138, 1977.
KB09	Kolda and Bader, tensor decompositions and applications, <i>SIAM Review</i> 51(3), 455–500, 2009.
O11	Oseledets, Tensor-Train decomposition, <i>SIAM Journal on Scientific Computing</i> 33(5), 2295–2317, 2011.
SB00	Sidiropoulos and Bro, uniqueness of multilinear decomposition of N-way arrays, <i>Journal of Chemometrics</i> 14(3), 229–239, 2000.
TB05	Tomasi and Bro, PARAFAC and missing values, <i>Chemometrics and Intelligent Laboratory Systems</i> 75(2), 163–180, 2005.
TB97	Trefethen and Bau, <i>Numerical Linear Algebra</i> , SIAM, 1997.

Final One-Page Memory Sheet

CP decomposition. A tensor is represented as a sum of rank-one outer products. CP rank is the smallest number of terms. CP is interpretable but can be ill-posed.

Unfoldings. Every CP rank- R tensor has every mode unfolding rank at most R . Therefore unfolding ranks are lower bounds for CP rank.

Arithmetic. Addition concatenates terms; scalar products use Gram matrices; Hadamard products can multiply ranks. Compression is usually needed after operations.

Geometry. Real tensors can have multiple typical ranks. Low CP-rank sets may not be closed. Border rank explains limits of low-rank sequences.

Uniqueness. Kruskal rank measures independence of every subset of columns. Kruskal-style inequalities give sufficient conditions for essential uniqueness up to scaling and permutation.

Algorithms. ALS updates one mode at a time through least squares and MTTKRP. Levenberg–Marquardt treats CP fitting as a coupled damped nonlinear least-squares problem. Compression from CP, Tucker, or TT formats uses contractions and Gram/Jacobian structure instead of full tensor expansion.